coping dynamics

EARLY-LIFE STRESS · FEAR CONDITIONING · BEHAVIOURAL SEQUENCING

Two ways of coping with the same early-life stress

Mice raised under limited bedding and nesting were fear-conditioned in adulthood, and their behaviour was sequenced with unsupervised keypoint-MoSeq rather than scored by an observer. The stressed group divides in two: a **vulnerable** subgroup whose sequences are more repetitive and less diverse than those of controls, and a **resilient** subgroup that does not differ from controls on any of those measures.

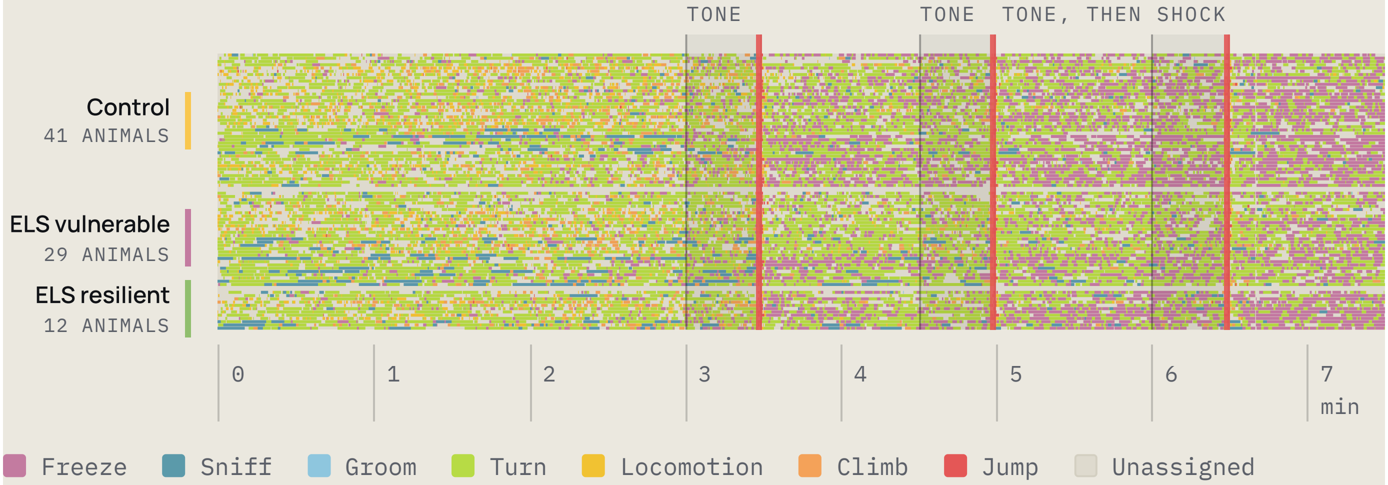
82 animals · 49 litters · 7 behavioural clusters · 250 ms bins · Code MIT

 · Rebuild SHA-256 verified

EVERY ANIMAL, EVERY 250 MS

The whole cohort as sequences

One row per mouse, one colour per behavioural cluster, seven and a half minutes of fear conditioning from left to right. Freezing rises after each tone, jumps mark each shock, and the three profiles are the three bands.



Each bin shows the cluster of its dominant syllable, read from `data/raw/syllable_usage_per_timebin_250ms.csv` with the cluster mapping and colours of Figure 6. Each 30 s tone ends in a 2 s foot shock. Rendered by `scripts/build_page_media.py`.

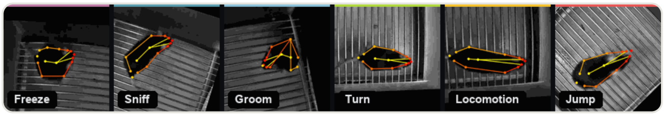
THE REPERTOIRE

Sequencing behaviour instead of scoring it

The stress response is usually reported as a set of labels assigned by an observer: fight, flight, freeze, fawn. The labels are categories imposed on a recording rather than quantities measured from it.

Keypoint-MoSeq segments pose tracking into short recurring motifs without supervision. Here the motifs map onto seven behavioural clusters, and each session becomes a sequence of those clusters at 250 ms resolution. The order of behaviour becomes measurable, not only the amount of each behaviour.

- **Freezing** — syllables 0, 28
- **Sniffing** — syllables 18, 20
- **Grooming** — syllable 24
- **Turn** — 8 syllables
- **Locomotion** — 7 syllables
- **Climbing** — geometry-derived
- **Jump** — 4 syllables



One representative keypoint-MoSeq syllable per behavioural cluster, egocentrically aligned, from hash-verified archived clips. Supplementary Video 1 plays on the project page, antonio-lozano.github.io/coping-dynamics-sequencing.

THE PARADIGM

Early-life stress, then fear conditioning in adulthood

Adversity in the first postnatal days, then a fear challenge at adulthood. The question is whether behaviour is organised differently when the threat returns, not only whether the stressed animals freeze more or less.

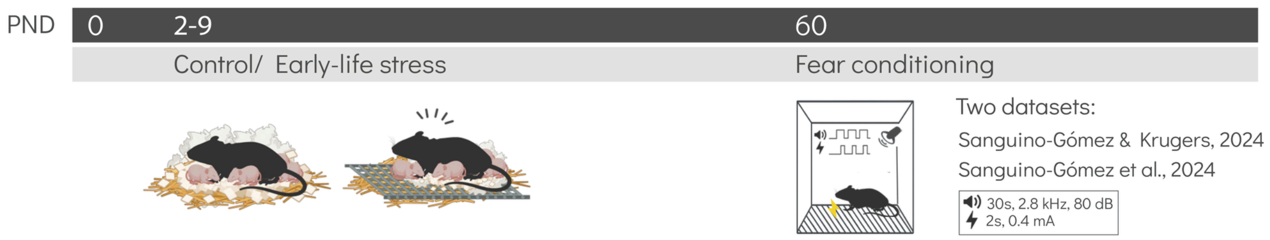


Figure 1A of the manuscript. Two cohorts, Sanguino-Gómez & Krugers 2024 and Sanguino-Gómez et al. 2024, are analysed together.

P2 – P9

Early-life stress

Limited bedding and nesting, the chronic-adversity model of Rice et al., 2008.

P60

Fear conditioning

Three 30 s tones at 2.8 kHz and 80 dB, each ending in a 2 s foot shock at 0.4 mA.

ANALYSIS

Behavioural sequencing

DeepLabCut pose tracking, then unsupervised keypoint-MoSeq syllables mapped onto seven behavioural clusters in 250 ms bins.

82
animals sequenced

49
litters, the clustering unit of every model

41 / 29 / 12
control, vulnerable, resilient

1,800
bins per session

FINDINGS

What the sequences show

Four figures, in the order the argument is built. Each regenerates from the bundled data with `scripts/run_all.py`, its plotted values are exported next to it, and each opens at full size and links to the vector PDF.

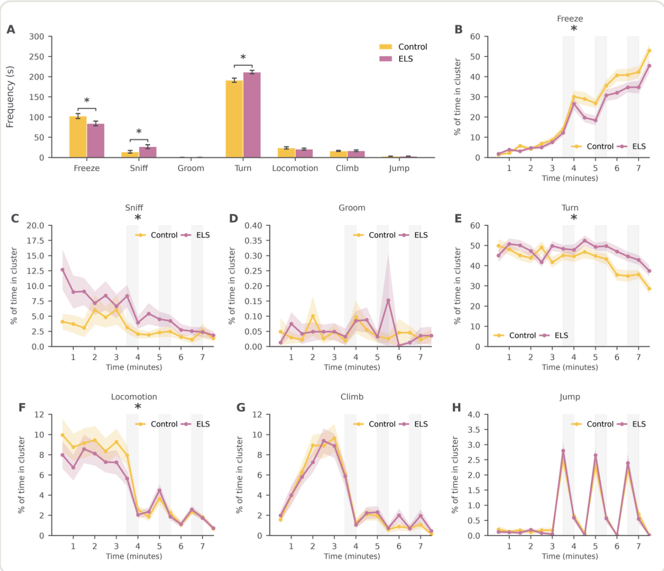


FIGURE 3

Behaviour frequencies and time courses differ between groups

Group differences appear both in how often each behaviour occurs and in how those frequencies change across the session, rather than as a constant offset between groups.

7

clusters, per animal

30 s

time-course bins

figure_source_data/figure3.csv

Vector PDF

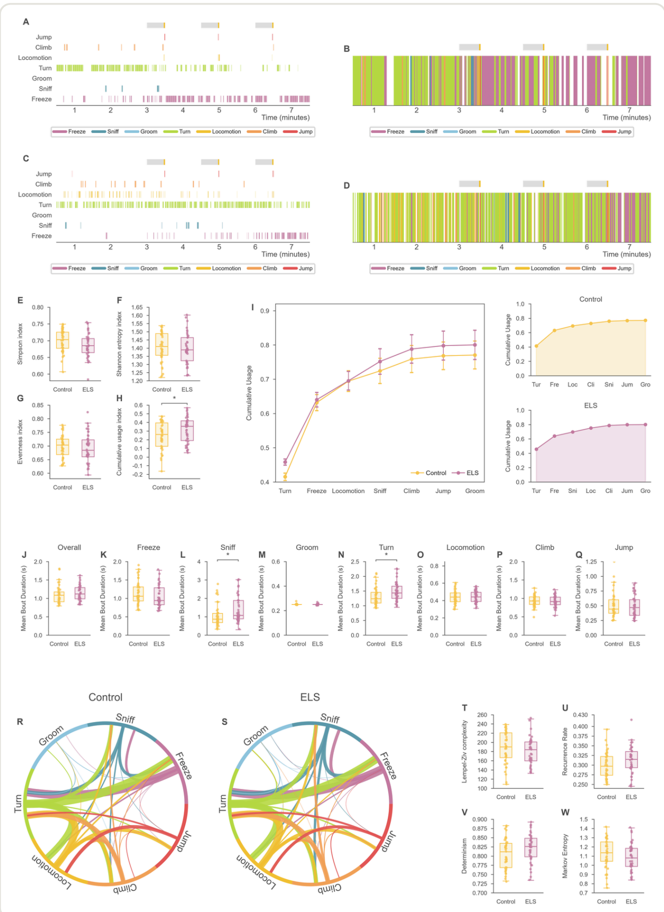


FIGURE 4

Stressed animals hold active behaviours longer

Sniffing and turn bouts last longer in the stressed group than in controls. Recurrence and determinism are higher too, but as trends that do not reach significance when all stressed animals are pooled.

+0.32 s

sniffing bouts, p = 0.029

+0.16 s

turn bouts, p = 0.020

statistics/stats_figure4_bouts_MixedLM.csv

Vector PDF

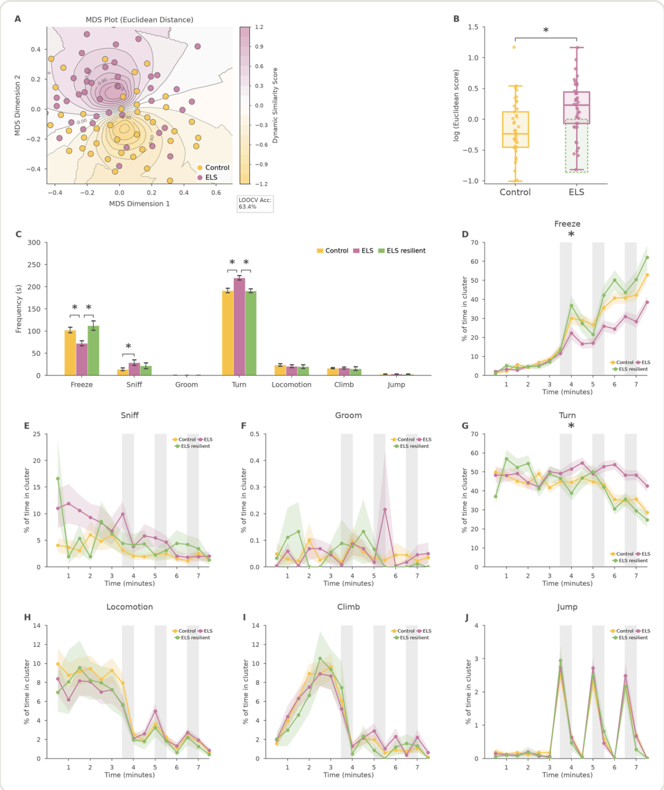


FIGURE 5

A dynamics score splits the stressed group in two

Each animal is scored by how close its behavioural dynamics sit to the control distribution. Clustering on that score separates the stressed animals into two profiles without using the outcome measures.

29

ELS vulnerable

12

ELS resilient

statistics/animal_dendrogram_clusters.csv

Vector PDF

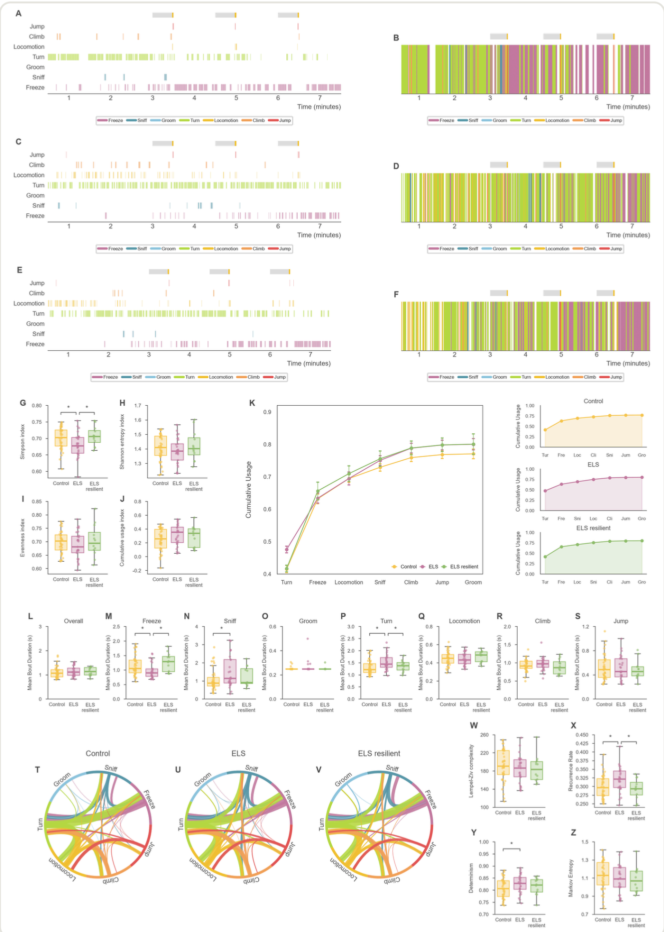


FIGURE 6

The split sharpens the group differences

Vulnerable animals return to the same states more often and use a narrower repertoire than controls. Resilient animals differ from the vulnerable group and not from controls.

+0.026

recurrence, q = 0.005

−0.022

Simpson diversity, q = 0.013

statistics/fig6_*_resilience_stats.csv

Vector PDF

TWO PROFILES

Vulnerable and resilient animals against controls

Pooled across all stressed animals, the sequence measures move in the expected direction but stop at trends. Split by coping profile, the picture is consistent: **on every measure that separates vulnerable animals from controls after FDR correction, resilient animals do not differ from controls.**

SEQUENCE MEASURE	ELS VULNERABLE VS CONTROL	ELS RESILIENT VS CONTROL
Recurrence rate how often the sequence revisits a state	+0.026 q = 0.005 DIFFERS	−0.008 q = 0.69 N.S.
Simpson diversity breadth of the repertoire in use	−0.022 q = 0.013 DIFFERS	+0.007 q = 0.77 N.S.
Freezing bout duration seconds	−0.18 s q = 0.028 DIFFERS	+0.08 s q = 0.73 N.S.
Sniffing bout duration seconds	+0.42 s q = 0.042 DIFFERS	+0.09 s q = 0.73 N.S.
Turn bout duration seconds	+0.22 s q = 0.028 DIFFERS	+0.02 s q = 0.84 N.S.

Mixed-effects coefficients grouped by animal, with Benjamini–Hochberg FDR correction within each metric family. The resilient-versus-vulnerable contrast is significant in the same direction for recurrence ($q = 0.005$) and Simpson diversity ($q = 0.010$). Every value is read from the shipped tables `statistics/fig6_transition_resilience_stats.csv`, `fig6_diversity_resilience_stats.csv` and `fig6_bout_resilience_stats.csv`, which the rebuild regenerates.

FOR READERS, REVIEWERS AND USERS

Where everything is

The repository is organised so that each claim in the manuscript can be traced to a figure, a source-data file, a fitted model and the data it was fitted on.

Repository

Code, data, figures and statistics in one clone. Code MIT, data CC BY 4.0.

Source data

The plotted values behind every panel, one CSV per figure, plus the Excel workbook.

`figure_source_data/ · report/raw_data.xlsx`

Statistics

Every fitted model with coefficients, tests, sample sizes and litter variance components.

`statistics/ · report/statistical_report.xlsx`

Data dictionary

Every bundled table, column by column, with units and provenance.

`docs/data_dictionary.md`

Videos and ethograms

The syllable atlas, and one ethogram and barcode for each of the 82 animals.

`supplementary_media/`

Companion classifier

Trained DeepLabCut networks, demo videos and tests, ready to run on new recordings.

`tools/behavior_dlc_classifier/`

Reproducibility

How the manifest, the checks and the CI keep every artifact byte-identical.

`REPRODUCIBILITY.md · MANIFEST.csv`

Availability

Data and code availability statements, licences and the archived release.

`DATA_AVAILABILITY.md · CODE_AVAILABILITY.md`

REPRODUCE

Rebuilding the analysis

Every figure, table, statistic and report on this page rebuilds from the data in the repository. There are no hidden inputs and no download step: a clone contains everything needed.

```
# install the locked environment
uv sync --locked

# verify the shipped artifacts against MANIFEST.csv
uv run python scripts/check_reproducibility.py

# rebuild everything; the tree ends clean
uv run python scripts/run_all.py
```

Hash-checked artifacts

`MANIFEST.csv` carries a SHA-256 for every artifact. A rebuild that changes one byte fails the check, on Windows and on Linux alike.

Statistics fitted once

Each model is fitted in one place and exported; `scripts/check_manuscript.py` pins every statistic quoted in the manuscript to the shipped report.

CITE

Sanguino-Gomez J, Güçlü U, Krugers HJ, Lozano A. Coping strategies dynamics and resilience profiles after early life stress revealed by behavioural sequencing. *bioRxiv*, 2025. doi:10.1101/2025.09.01.673507. The code and data are cited separately below; an archived, DOI-minted release accompanies the manuscript at submission.

```
@article{sanguino_gomez_2025_coping,
  author  = {Sanguino-Gomez, Jeniffer and Güçlü, Umut and Krugers, Harm J. and Lozano, Antonio},
  title   = {Coping strategies dynamics and resilience profiles after early life stress revealed by behavioral sequencing},
  journal = {bioRxiv},
  year    = {2025},
  doi     = {10.1101/2025.09.01.673507}
}

@software{sanguino_gomez_2026_coping,
  author  = {Sanguino Gomez, Jeniffer and Lozano, Antonio},
  title   = {Coping Dynamics Sequencing},
  year    = {2026},
  version = {1.0.0},
  url     = {https://github.com/antonio-lozano/coping-dynamics-sequencing},
  license = {MIT}
}
```

Coping Dynamics Sequencing — Jeniffer Sanguino-Gómez and colleagues, Brain Plasticity group, SILS-CNS, University of Amsterdam.

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 Citation metadata lives in `CITATION.cff`; GitHub's "Cite this repository" reads it. Built with DeepLabCut and keypoint-MoSeq.

github.com/antonio-lozano/coping-dynamics-sequencing